PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

# DDH

## Who gets it

- Risk factors: **breech** (strongest), female, first-born, family history, oligohydramnios, tight swaddling with hips extended. **Left hip** in about 60%.
- Associated "packaging" conditions: torticollis, metatarsus adductus, congenital knee dislocation.

## Examination by age

- **Newborn: Barlow** (adduct + push posteriorly: dislocatable hip) and **Ortolani** (abduct + lift trochanter: dislocated hip reduces with a clunk).
- **After about 3 months:** these become negative. **Limited abduction** is the most reliable sign; **Galeazzi** (unilateral, knee lower); Klisic line.
- **Walking child:** Trendelenburg limp (unilateral), waddling gait and hyperlordosis (bilateral).

## Imaging

- **Ultrasound** until the femoral head ossifies (about 4-6 months). **Graf alpha angle** of 60 degrees or more is normal (type I). Type IIa (50-59 degrees, under 3 months) is physiological immaturity: re-scan. Lower angles and types III/IV need treatment. Femoral head coverage under 50% is abnormal.
- **Radiograph** after 4-6 months: Hilgenreiner and Perkin lines (head should lie in the inferomedial quadrant), broken **Shenton line**, raised **acetabular index**, delayed ossific nucleus. In older

children and adults, a **centre-edge angle** under 20 degrees means dysplasia.

## Treatment by age

| Age | Treatment |
|---|---|
| 0-6 months | **Pavlik harness**: hips flexed 90-100 degrees, abduction not forced. Confirm reduction on ultrasound; stop if not reduced in 3-4 weeks |
| 6-18 months (or Pavlik failure) | **Closed reduction** under anaesthesia with arthrogram, adductor tenotomy, spica in the "human position" (about 100 degrees flexion, abduction within the Ramsey safe zone) |
| over 18 months | **Open reduction**, often with femoral shortening and pelvic osteotomy |
| beyond about 8 years unilateral (6 bilateral) | Usually leave a high dislocation unreduced |

- **Pavlik complications:** femoral nerve palsy (excess flexion), inferior dislocation, AVN (forced abduction), "Pavlik disease" (posterior wall erosion from persisting with a non-reduced hip).
- **Pelvic osteotomies: Salter** (redirection, 18 months-6 years, needs a concentric reduction); **Pemberton** (reshaping, capacious acetabulum, open triradiate cartilage); **Dega** (posterior deficiency, neuromuscular hips); after triradiate closure, **Ganz periacetabular osteotomy**.

## Blocks to reduction

- **Extra-articular:** iliopsoas (hourglass capsule), adductor longus.

- **Intra-articular:** pulvinar, hypertrophied ligamentum teres, transverse acetabular ligament, inverted labrum (limbus).
- The **medial (Ludloff) approach** in infants endangers the medial femoral circumflex artery.

> Exam pearl: **AVN of the femoral head** is the most important complication of treatment, not of the disease itself (Kalamchi-MacEwen classification).

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.